

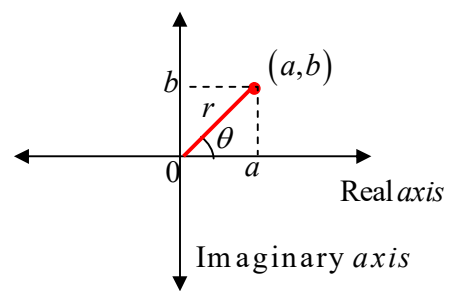
Complex Analysis

5.3 Singularity

Poles: For a rational function $f(z)$ in reduced form, the poles are the values of z where the denominator is equal to zero; or in other words, the points where the rational function is not defined are known as poles.

Like if a function is given as $f(z)$ about $z - z_0$ having some finite power as $(z - z_0)^{-n}$, where n is an integer. In this case, the singularity is called a pole and is more specifically identified as a pole of order n . A pole of order one is called a simple pole.

And if the expansion of $f(z)$ about $z - z_0$ will continue to negatively infinite powers of $z - z_0$. This case is not referred to as a “pole of infinite order,” but is called an essential singularity.



⇒ Find the roots of the following functions.

(a) $f(x) = x^2 - 1$

Solution: $x^2 = 1 \Rightarrow x^2 = e^{i2n\pi} \Rightarrow x = e^{in\pi}$

$x_0 = 1, x_1 = -1$

(b) $f(x) = x^2 + 1$

Solution: $f(x) = 0 \Rightarrow x^2 = -1 \Rightarrow x^2 = e^{i[2n+1]\pi}$

$\Rightarrow x = e^{i(2n+1)\pi/2} \Rightarrow x_0 = i, x_1 = -i$

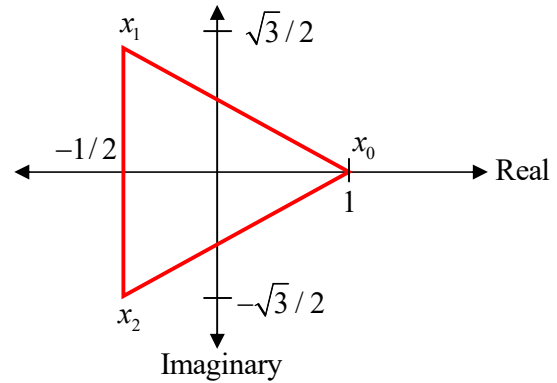
(c) $f(x) = x^3 - 1$

Solution: $x^3 - 1 = 0 \Rightarrow x^3 = 1$

$x = 1 = e^{i2n\pi/3}, x_0 = 1 (n=0)$

$x_1 = e^{i2\pi/3} = -\frac{1}{2} + \frac{i\sqrt{3}}{2} (n=1)$

$x_2 = e^{i4\pi/3} = -\frac{1}{2} - i\frac{\sqrt{3}}{2} = -\frac{(1+i\sqrt{3})}{2} (n=2)$



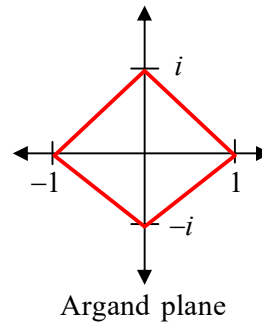
(d) $f(x) = x^4 - 1$

Solution: $f(x) = x^4 - 1 = 0$

$x^4 = 1 = e^{i2n\pi}, x = e^{\frac{i2n\pi}{4}} = e^{\frac{in\pi}{2}}$

$x_0 = 1 (n=0), x_1 = i (n=1), x_2 = e^{i\pi} = -1 (n=2)$

& $x_3 = e^{i3\pi/2} = -i (n=3)$



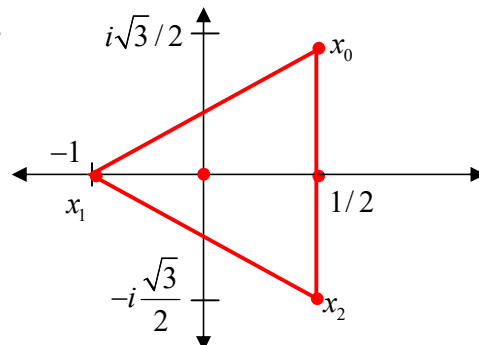
(e) $f(x) = x^3 + 1$

Solution: $f(x) = 0 \Rightarrow x^3 = -1 = e^{i(2n+1)\pi}, x = e^{i(2n+1)\pi/3}$

$x_0 = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} = \frac{1}{2} + i\frac{\sqrt{3}}{2}$

$x_1 = e^{i\pi} = -1, x_2 = e^{i5\pi/3} = \cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3}$

$x_2 = \frac{1}{2} - i\frac{\sqrt{3}}{2}$



(f) $f(x) = x^4 + 1$

Solution: $f(x)=0 \Rightarrow x^4 = -1, \quad x^4 = e^{i(2n+1)\pi} \Rightarrow x = e^{i(2n+1)\pi/4}$

$$x_0 = e^{i\pi/4} = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}}$$

$$x_1 = e^{i3\pi/4} = \cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} = -\frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}}$$

$$x_2 = e^{i5\pi/4} = \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} = -\frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}}$$

$$x_3 = e^{i7\pi/4} = \cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4} = \frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}}$$

$\Rightarrow$ Find the poles of the following functions.

(a) $f(x) = \frac{1}{x^2 - 1}$

Solution: $x^2 - 1 = 0, \quad x_0 = 1 \text{ \& } x_1 = -1$ are poles.

(b) $f(x) = \frac{1}{x^2 + 1}$

Solution: $x^2 + 1 = 0 \Rightarrow x_0 = i \text{ \& } x_1 = -i$ are the poles.

(c) $f(x) = \frac{1}{x^4 + 1}$

Solution: $x^4 + 1 = 0 \Rightarrow x^4 = -1 = e^{i(2n+1)\pi} \Rightarrow x = e^{i(2n+1)\pi/4}$

$$x_0 = \frac{1+i}{\sqrt{2}}, \quad x_1 = -\frac{(1-i)}{\sqrt{2}}$$

$$x_2 = -\frac{(1+i)}{\sqrt{2}}, \quad x_3 = \frac{1-i}{\sqrt{2}}$$

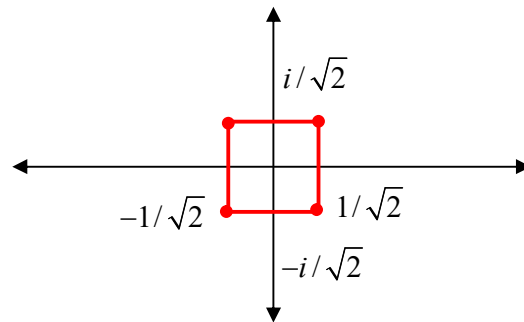
(d) $f(x) = \frac{1}{x^6 + 1}$

Solution: $x^6 + 1 = 0 \Rightarrow x^6 = -1 \Rightarrow x^6 = e^{i(2n+1)\pi} \Rightarrow x = e^{i(2n+1)\pi/6}$

$$\Rightarrow x_0 = e^{i\pi/6} = \cos \frac{\pi}{6} + i \sin \frac{\pi}{6} = \frac{\sqrt{3}}{2} + i \frac{1}{2}$$

$$x_1 = e^{i3\pi/6} = e^{i\pi/2} = i$$

$$x_2 = e^{i5\pi/6} = \cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} = -\frac{\sqrt{3}}{2} + \frac{i}{2}$$



$$x_3 = e^{i7\pi/6} = \cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6} = -\frac{\sqrt{3}}{2} - \frac{i}{2}$$

$$x_4 = e^{i9\pi/6} = \cos \frac{9\pi}{6} + i \sin \frac{9\pi}{6} = -i$$

$$x_5 = e^{i11\pi/6} = \cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6} = \frac{\sqrt{3}}{2} - \frac{i}{2}$$

So $(x_0, x_1, x_2, x_3, x_4, x_5)$ are the poles.

(e) $f(x) = \frac{1}{x^4 + 4}$

Solution: $x^4 + 4 = 0 \quad x^4 = -4 \Rightarrow x^4 = 4(-1) = 4e^{i(2n+1)\pi}$

$$x = 4^{1/4} \cdot e^{i(2n+1)\pi/4} = \sqrt{2} \cdot e^{i(2n+1)\pi/4}$$

$$x_0 = \sqrt{2} \cdot e^{i\pi/4} = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) = 1 + i$$

$$x_1 = \sqrt{2} \cdot e^{i3\pi/4} = \sqrt{2} \left[\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right] = -1 + i$$

$$x_2 = \sqrt{2} \cdot e^{i5\pi/4} = \sqrt{2} \left[\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right] = -1 - i$$

$$x_3 = \sqrt{2} \cdot e^{i7\pi/4} = \sqrt{2} \left[\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4} \right] = 1 - i$$

So the poles are x_0, x_1, x_2, x_3 .

